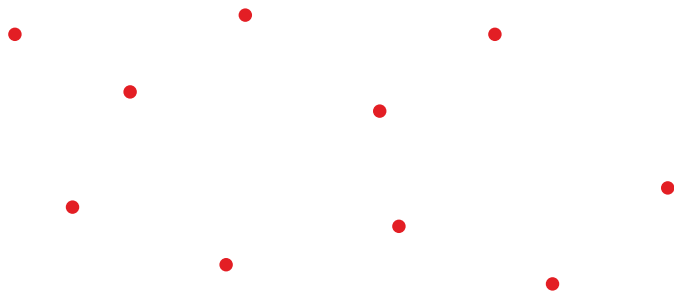


OMNIKON

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MISSION_SEARCH: ACTIVE

ONE MISSION. BUILD THE IMPOSSIBLE.

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TEAM NAME

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MEMBERS

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THEME

Warehouse Inventory Optimization for Small Logistics Providers

PROBLEM STATEMENT

Warehouse Inventory Optimization for Small Logistics Providers

Small and medium logistics providers often lack sophisticated tools for inventory optimization available to larger players. Create a solution tailored to their needs.

PROBLEM UNDERSTANDING

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Small logistics providers and 3PL operators manage multi-client inventory with limited budgets — unable to afford enterprise WMS platforms like SAP or Oracle, they fall back on spreadsheets, physical counts, and gut-feel reordering.

■ Stockouts

Poor visibility into real stock levels damages client trust and revenue.

■ Overstocking & Dead Stock

Capital and space tied up in slow-moving or obsolete SKUs.

■ Inventory Inaccuracy

Mismatches between recorded and physical stock from manual entry.

■ Inefficient Space & Labor

No data-driven basis for slotting, reordering, or prioritization.

ROOT CAUSE

A "5 Whys" analysis traces every symptom to one structural gap: small operators lack affordable, simple decision-support tools — the market offers only expensive enterprise WMS or unstructured spreadsheets, nothing built for the small-operator scale in between.

PROBLEM STATEMENT (CONCISE)

How might we help small logistics providers gain real-time inventory visibility and data-driven reorder decisions — without the cost or complexity of enterprise systems?

PROPOSED SOLUTION

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A lightweight, mobile-first inventory visibility platform — not a full enterprise WMS. Most small operators don't need complex warehouse orchestration; they need fast counting, clear trends, and timely alerts.

DESIGN PHILOSOPHY Every feature is scoped against one question — does this help a non-technical warehouse manager make a faster, better decision? Deep ERP integrations and elaborate configuration are later-stage stretch goals, not core requirements.

WHY IT ADDRESSES THE PROBLEM

1 Closes the mid-market gap

Sits between costly enterprise WMS and unstructured spreadsheets.

2 No hardware barrier

Browser-based barcode scanning works on any phone — no app install.

3 Decision support, not just data

Reorder points, dead-stock flags, and ABC tiers turn data into action.

Innovation: combines mobile scanning, automated decision rules, and applied ML forecasting in a single low-cost tool purpose-built for the small-operator scale — a segment enterprise WMS vendors ignore.

SYSTEM ARCHITECTURE

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Layered architecture — from device input through to decision-ready dashboards.



Data flows bottom-up (scan → API → database) and top-down for alerts (reorder points → dashboard notifications).

KEY FEATURES

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1

Mobile Barcode Scanning

Camera or handheld-scanner input logs stock received, picked, or shipped — no dedicated hardware or app install.

2

Real-Time Inventory Visibility

Live dashboard showing current stock levels, recent movement, and top/bottom performing SKUs.

3

Automated Reorder Alerts

System-calculated reorder points, based on usage, lead time, and safety stock, trigger low-stock notifications.

4

Dead Stock Detection

Rule-based flagging of SKUs with no movement over a configurable window, with suggested actions.

5

ABC Inventory Classification

Automatic segmentation of SKUs by value and velocity, so operators focus attention where it matters.

6

Focused KPI Dashboard

Turnover ratio, stockout rate, inventory accuracy, and days on hand — not an overwhelming report suite.

1

Technical Feasibility

Uses React, Python, PostgreSQL, and barcode scanning.

2

Operational Feasibility

Mobile scanning makes warehouse tracking simple and fast.

3

Financial Viability

Open-source tools and cloud hosting keep costs low.

4

Challenges and Risks

Risks include inaccurate data, poor connectivity, barcode errors, and staff resistance.

5

Mitigation Strategies

Use CSV validation, manual entry, audit logs, backups, user training, and phased implementation.

6

Scalability and Impact

The platform can expand from one warehouse to multi-client 3PL operations while reducing stockouts and overstocking.

IMPACT AND BENEFITS

1 Affordable inventory management for small logistics providers.

2 Fewer stockouts, overstocking, and dead-stock losses.

3 Faster barcode-based warehouse operations.

4 Lower storage, labor, and emergency-purchase costs.

5 Better customer trust through accurate inventory visibility.

6 Reduced packaging waste and unnecessary transportation.

TECHNOLOGY STACK

LAYER	TECHNOLOGY	WHY
Frontend	React (Next.js), Tailwind CSS	Responsive, mobile-first UI
Backend / API	Node.js (Express) or Python (FastAPI)	REST API; ready for Shopify/WooCommerce sync
Database	PostgreSQL	Relational integrity, multi-tenant isolation
Barcode Scanning	QuaggaJS / ZXing (browser camera)	No dedicated hardware or app install
Analytics / Forecasting	Python — pandas, scikit-learn, statsmodels	ABC classification, demand forecasting
Visualization	Recharts / Chart.js	KPI dashboards and trend charts
Hosting	Vercel / Render	Minimal DevOps, SMB-scale cost
Authentication	JWT with role-based access control	Multi-client / multi-user access (Phase 2)

IMPLEMENTATION PLAN

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A phased roadmap delivers a working, demo-ready product after Phase 1 — each later phase adds depth rather than gating usability.

01

PHASE 1

Core Inventory & Scanning

Single-warehouse inventory tracking with scan-based stock movement

KEY DELIVERABLES

- Product / SKU database
- Barcode scan-in / scan-out
- Reorder point engine
- Low-stock alerts

02

PHASE 2

Multi-Client Dashboard

Support for multiple client accounts within one warehouse (3PL use case)

KEY DELIVERABLES

- Multi-tenant data model
- Per-client views
- Role-based access control

03

PHASE 3

Analytics & Forecasting

Predictive and classification features layered on top of Phase 1–2 data

KEY DELIVERABLES

- ABC analysis automation
- Demand forecasting (moving average / ML)
- Dead-stock detection engine

RESEARCH AND REFERENCES

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Reviewer Notes

This submission describes a proposed, lightweight MVP. Barcode hardware, historical datasets, forecast accuracy, deployment metrics, and user-validation results are not yet available; therefore, the results section defines target outcomes and validation measures rather than claiming measured performance. Items marked Phase 2 or Phase 3 are future extensions beyond the core demo.